

Department of Software Engineering

Mehran University of Engineering and Technology, Jamshoro

Course: Web Engineering

Instructor	Dr. Din Muhammad	Assignment Type	Complex Engineering Problem
Semester	7 th	Year	4 th (Final)
Submission Deadline	11th May (In-Person) OR If a group is unable to present on 11th May, they may present online any time up to 15th May		
Assessment Score	15 Marks		

Complex Engineering Problem - Characteristics

Code	Characteristic	Justification
CEP - A1	Depth of Knowledge Required	Requires knowledge of web development, database design, PHP & MySQL, and session-based authentication.
CEP - A2	Range of Conflicting Requirements	Balancing ease of use for students vs. security of job data, and simplicity vs. feature richness.
CEP - A3	Depth of Analysis Required	Needs analysis of user roles, data flow between employers and students, and filtering logic for job search.
CEP - A4	Infrequently Encountered Issues	Managing multiple application statuses, preventing duplicate applications, and handling concurrent form submissions.
CEP - A5	Beyond Standard Practice	Integrating role-based dashboards and application status tracking goes beyond basic web development exercises.
CEP - A6	Diverse Stakeholder Groups	Students, employers, and admins each have different goals and system access requirements.
CEP - A7	Interdependence (Sub-problems)	Job posting, application management, user authentication, and admin moderation are all connected sub-systems.
CEP - A8	Significant Consequences	Poor design can result in missed job opportunities for students or exposure of private employer data.
CEP - A9	Judgement in Decision Making	Requires decisions on database schema design, access control logic, and UI layout for different user roles.

Problem Description

Student Job Portal

Background:

Many students struggle to find part-time jobs or internships while universities and employers have no simple platform to connect with them. A Student Job Portal will allow employers to post job/internship listings and students to browse, apply, and track their applications, all through a simple web interface.

Features to Implement:

- User registration & login for two roles: Student and Employer
- Employer can post, edit, and delete job/internship listings
- Students can browse job listings and apply with a short cover note
- Students can view and track the status of their applications (Pending / Accepted / Rejected)
- Employers can view applications for their posted jobs and update application status
- Admin panel to manage users and remove fake/spam job posts

Tasks

1. Responsive Front-End Design (CEP – A1, A3, A9)

Design a clean, responsive UI using HTML5, CSS, and Bootstrap. Build separate views for Student, Employer, and Admin roles. Apply form validation on registration and job application forms.

2. User Registration & Login (CEP – A1, A3)

Implement registration and login for Students and Employers using PHP & MySQL. Use session management to keep users logged in and restrict pages based on their role.

3. Job Listing System (CEP – A1, A7)

Allow employers to post new jobs with: job title, description, type (part-time/internship), deadline, and location. Store listings in MySQL and display them to students with search and filter options.

4. Application System (CEP – A2, A8)

Allow students to apply for jobs with a short cover note. Prevent duplicate applications for the same job. Let employers view received applications and mark them as Accepted or Rejected.

5. Application Status Tracking (CEP – A3, A9)

Show students a dashboard listing all their submitted applications with current status (Pending, Accepted, Rejected). Update status in real-time when employer takes action.

6. Admin Panel (CEP – A3, A9)

Allow the admin to view all users and job posts. Admin can delete spam/fake job listings and ban/unban user accounts.

Constraints

- Only registered users can post jobs or submit applications.
- A student cannot apply to the same job more than once.
- Only the employer who posted a job can update or delete it.
- Admin can remove any job post or user account.
- Job listings must include a deadline; expired listings should be hidden from students.
- Project must be completed in a group of maximum 3 students.

Research Work

- Study existing job portals (e.g., LinkedIn, Indeed, Rozee.pk) for UI/UX reference.
- Research role-based access control (RBAC) in PHP web applications.
- Explore common web security vulnerabilities (SQL Injection, XSS) and how to prevent them.

Deliverables

- A working web prototype with all core features (job posting, applying, status tracking, admin panel).
- Project report: database schema, system architecture, UI screenshots, and challenges faced.
- Test case screenshots showing registration, job posting, application flow, and admin actions.

Notes:

Keep it simple. Focus on core features only. Use Bootstrap for styling. Use PHP & MySQL or any for the back-end. No need for complex frameworks or real-time features.

Assignment Presentation Instructions

All groups are required to present their fully functional (running) website project on their own laptops.

- **Primary Presentation Date:** 11th May (In-Person)
- **Alternate Option:** If a group is unable to present on 11th May, they may present online any time up to 15th May (no extensions beyond this date).

During the presentation:

- The project must be demonstrated live (not slides or screenshots).
- Every group member should be prepared to explain their contribution.
- **Technical and coding-related questions** will be asked based on your project (including implementation, logic, tools used, and problem-solving approaches).

Important Notes:

- Make sure your project runs properly without errors.
- Bring all required setup (local server, database, files, etc.).
- Marks will be affected if the project is not functional or if students are unable to answer questions.

Rubrics (Each Rubric 3 Marks)

Rubric	Unacceptable (0)	Poor (1)	Acceptable (2)	Proficient (3)
R1 UI/UX Design & Responsiveness				
R2 User Registration & Role Management				
R3 Job Listing & Search System				
R4 Application & Status Tracking, Admin Panel & Security				
R5 Viva Voce				
Total Marks				/ 15